Method for Analyzing Heat Shock Protein Content in Brain Tissue Using Machine Learning Based on Raman Spectra

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Raman scattering is a powerful non-destructive method for analyzing the composition of biological tissues. Raman spectra contain information about the molecular composition of a sample and make it possible to detect biochemical changes accompanying the development of various diseases. Heat shock proteins (HSPs) represent a highly promising group of compounds that may find applications in clinical medicine, including neurology, pharmacology, and oncology.

Of particular interest for neurobiology is the heat shock protein with a molecular weight of 70 kDa (HSP70). It is capable of binding numerous pathological proteins with misfolded conformations that arise during the progression of neurodegenerative diseases (e.g., Alzheimer's disease) and promotes their refolding. Studying the content of these proteins in different regions of the brain of transgenic mice expressing elevated levels of exogenous and endogenous HSP will allow for a more precise understanding of their concentration and distribution within the structures of the central nervous system.

However, the practical application of Raman spectroscopy in medical diagnostics faces a number of challenges, including high data dimensionality, the presence of noise, fluorescence background, and the complexity of spectral interpretation. The aim of this study is to develop a method for determining and analyzing the content of 70 kDa heat shock proteins in brain tissue sections of transgenic mice using Raman spectroscopy.

In this study, a provided secondary dataset containing Raman spectra of brain tissue sections was analyzed. The initial dataset included measurements from three experimental groups of animals: wild-type mice (control group) and two lines of transgenic mice with an integrated genetic cassette enabling overexpression of the human heat shock protein (HSP70) in the cytosol (endogenous expression) and in the interstitial space (exogenous expression).

According to the metadata of the provided dataset, Raman spectra acquisition was performed using a confocal Raman microscope (Renishaw inVia Qontor). Measurements were carried out in the primary motor cortex and striatum of the brain. The acquisition parameters included the use of an excitation laser with a wavelength of 633 nm.

The original dataset is characterized by the presence of measurements across different frequency ranges, the selection of which is determined by the central pixel. The shift of the central pixel is achieved by mechanical rotation of the diffraction grating. To avoid geometric distortions, spectra with different central pixels were not merged into a single dataset and were instead analyzed separately.

To suppress noise, correct background signals, and prepare the spectral data for training a classification model, the following algorithm was implemented:

1. Interpolation. To bring the data to a unified dimensionality and eliminate shifts along the wavenumber axis, the original intensity values were projected onto a uniform one-dimensional grid. This ensured model robustness when working with spectra having different discretization steps.
2. Baseline correction. Biological tissues exhibit intrinsic fluorescence, which creates a nonlinear low-frequency background. To remove it, the Asymmetric Least Squares (ALS) method was applied. This approach allowed iterative approximation and subtraction of the fluorescence component without distorting the shape and intensity of sharp Raman peaks.
3. Noise filtering. The Savitzky–Golay method was used to suppress high-frequency stochastic noise. This smoothing algorithm is well-suited for spectroscopy, as it preserves information about the width and height of spectral features.
4. Normalization. To compensate for instrumental fluctuations (e.g., laser power), L2 normalization (vector normalization) was applied, bringing each spectrum to unit length.
5. Differentiation. To resolve overlapping spectral bands and remove possible residual linear trends, the first and second derivatives of the normalized spectra were computed.

Based on the preprocessed data, a baseline classification model was built using the CatBoost gradient boosting algorithm. Training was performed independently for two frequency ranges: low-frequency (centered at 1500 cm^{-1}) and high-frequency (centered at 2900 cm^{-1}). Model performance was evaluated at the level of spectral

maps using two aggregation strategies: majority voting and probability averaging. The classification results obtained using the voting method are presented in the table below.

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Table 1. Classification metrics using the voting method (Voting)

Class	**Precision**	**Recall**	**F1-score**
Control 1500 cm^{-1}	0.29	0.23	0.25
Exo 1500 cm^{-1}	0.76	0.74	0.75
Endo 1500 cm^{-1}	0.24	0.31	0.27
Control 2900 cm^{-1}	0.40	0.35	0.37
Exo 2900 cm^{-1}	0.50	0.62	0.55
Endo 2900 cm^{-1}	0.52	0.44	0.48

When switching to the strategy of probability averaging over the entire area of the spectral map, an overall improvement in the predictive performance of the model trained on low-frequency spectra was observed. The corresponding metrics are presented in the table below.

Table 2. Classification metrics using the averaging method (Averaging)

Class	**Precision**	**Recall**	**F1-score**
Control 1500 cm^{-1}	0.36	0.35	0.35
Exo 1500 cm^{-1}	0.78	0.83	0.80
Endo 1500 cm^{-1}	0.24	0.22	0.23
Control 2900 cm^{-1}	0.38	0.38	0.38
Exo 2900 cm^{-1}	0.50	0.62	0.55
Endo 2900 cm^{-1}	0.48	0.36	0.41

The analysis of the obtained results showed that the group with exogenous expression of the HSP70 protein is the most clearly distinguishable from the other classes, achieving an F1-score of 0.80. At the same time, the control and endogenous groups demonstrate a high degree of spectral similarity, which makes their differentiation using baseline methods difficult. The advantage of the averaging method indicates the importance of accounting for the global spectral profile of the map to mitigate local noise effects. The identified patterns served as the basis for developing a more complex model architecture that integrates data from different frequency ranges.

To overcome the limitations of baseline algorithms and effectively account for the spatial heterogeneity of the data, a neural network solution was developed based on the paradigms of weakly supervised learning and Multi-Instance Learning (MIL). Within this approach, a spectral map is treated as a set of individual spectra, and training is performed exclusively using map-level labels. The model architecture includes a compact multiscale one-dimensional convolutional neural network (multiscale 1D CNN) as an encoder for feature extraction from individual spectra, as well as a Gated Attention MIL module for aggregating the resulting vector representations into a final prediction for the entire map. Additionally, a specialized module was implemented to enable inference at the level of individual measurements using an ensemble of models and averaging of predicted probabilities.

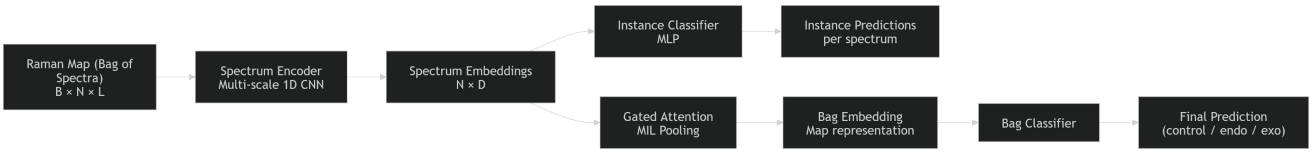

Figure 1. Diagram of the proposed neural network model

The model training process was divided into two stages. During the warmup stage, each individual spectrum was temporarily assigned the label of the entire map, after which the training proceeded to joint MIL learning with aggregation at the map level. The optimization included the computation of a loss function at the map level and pseudo-instance supervision for the most informative spectra. The validation strategy followed the Leave-One-Group-Out (LOGO) principle with splitting by groups of spectral maps, which strictly prevented data leakage

between measurements belonging to the same biological section. A critically important engineering decision was to exclude metadata from the set of input features, which made it possible to avoid spurious correlations related to the specifics of data splitting. To improve training stability, traditional batch normalization was replaced with group normalization (GroupNorm).

The optimized neural network architecture demonstrated improved classification performance compared to earlier versions, which experienced significant difficulties in recognizing the group with endogenous expression. The final model, when evaluated across all brain regions, achieved a macro-averaged F1-score of 0.5308 and an overall accuracy of 0.5424. For the control group, the F1-score and recall were 0.5063 and 0.5000, respectively, while for the endogenous expression group they were 0.4118 and 0.3889. An important advantage of the proposed solution is its interpretability: when analyzing an individual spectrum, the model outputs not only class probabilities but also highlights informative Raman peaks using the Integrated Gradients method, producing an attribution curve along the spectral axis to explain its predictions.

When evaluating the obtained results, it is important to consider the inherent limitations of the provided dataset. In particular, one of the experimental groups contained spectra obtained exclusively from the cerebellum. Due to the strict data separation under the LOGO cross-validation scheme, one of the evaluation folds effectively represented a structural domain shift scenario. This results in a certain heterogeneity in model performance across different tissue types, reflecting not only the complexity of spectral classification but also the effect of feature space transfer between different anatomical regions of the central nervous system.

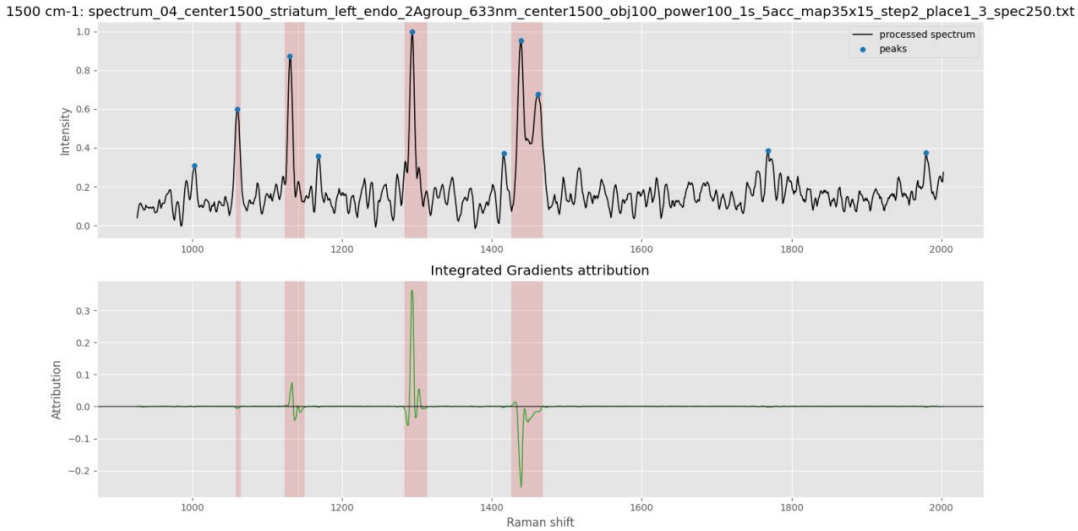

Figure 2. Attribution map of spectral features for spectra centered at 1500 cm^{-1}

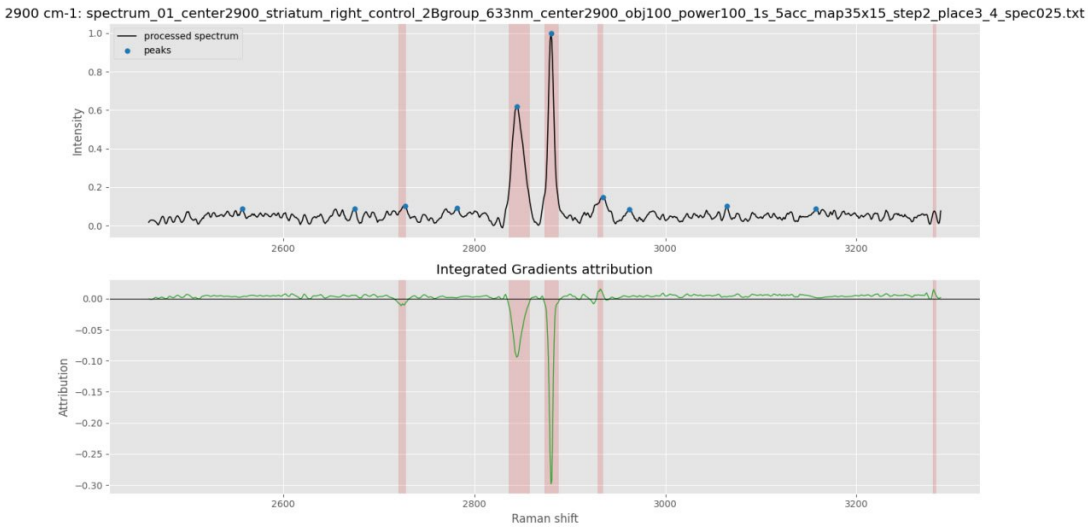

Figure 3. Attribution map of spectral features for spectra centered at 2900 cm^{-1}

Thus, the developed neural network pipeline based on a multi-instance learning architecture with an attention mechanism not only improves the classification accuracy of heterogeneous spectral data but also enables the extraction of biologically meaningful information. The spectral importance profiles obtained using gradient-

based attribution allow localization of specific frequency ranges that contribute most to the differentiation of control, endogenous, and exogenous groups. The informative Raman peaks identified by the model can serve as a starting point for further biochemical analysis of conformational changes and the distribution of heat shock proteins. This opens up broad prospects for studying the molecular mechanisms of the neuroprotective effects of HSP70 and for discovering new optical biomarkers of pathological processes in the central nervous system.